An Unsupervised Adaptive PINN-NTK Framework for Transport Phenomena in Hybrid Nanofluid Flow

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Introduction

- The Non-Newtonian model is a widely used non-Newtonian fluid model that captures shear-thinning behavior, with viscosity varying continuously with the shear rate [1].
- In rotating channel flow, the Non-Newtonian fluid formulation leads to a coupled nonlinear ODE system with strong nonlinearity and boundary constraints [2].
- Physics-Informed Neural Networks (PINNs) provide a mesh-free framework for solving such systems by embedding governing equations and boundary conditions into training [3].
- To improve the loss imbalance and slow convergence in standard PINNs, this work applies PINN-NTK [4], which uses adaptive weighting to enhance training stability and accuracy.

Keywords: Non-Newtonian Fluid; PINN; Neural Tangent Kernel; Adaptive technique

Mathematical Modeling

Model Description

- In rotating channel flow, the Non-Newtonian constitutive relation introduces strong nonlinearity into the governing equations, which significantly increases the difficulty of mathematical analysis, numerical approximation, and stable computation.
- Owing to its ability to represent realistic transport behavior, the Non-Newtonian model provides an important framework for studying velocity, temperature, and concentration fields.
- The model also incorporates a porous medium and a CMC–water hybrid nanofluid with dispersed Al_2O_3 and Fe_2O_3 nanoparticles.

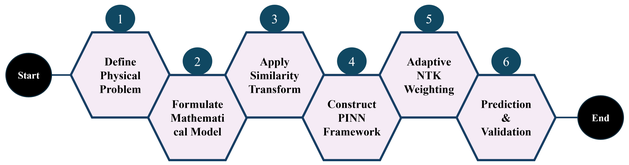


Figure 1. Research Framework

Dimensionless nonlinear set of ordinary differential equations:

$$\frac{P}{\Delta_2}+\frac{\Delta_1}{\Delta_2}\left[\left\{1+\left(We\frac{du}{d\xi}\right)^2\right\}^{\frac{n-1}{2}}\right]\frac{d^2u}{d\xi^2}+\frac{\Delta_1}{\Delta_2}(n-1)We\left[1+\left(We\frac{du}{d\xi}\right)^2\right]^{\frac{n-3}{2}}\left(\frac{du}{d\xi}\right)^2\frac{d^2u}{d\xi^2} - \frac{\Delta_1 Da}{\Delta_2}u\left[1+\left(We\frac{du}{d\xi}\right)^2\right]^{\frac{n-1}{2}} - D_f u^2 - 2\Omega\omega = 0 \tag{1}$$

$$\frac{P}{\Delta_2}+\frac{\Delta_1}{\Delta_2}\left[\left\{1+\left(We\frac{d\omega}{d\xi}\right)^2\right\}^{\frac{n-1}{2}}\frac{d^2\omega}{d\xi^2}\right]+\frac{\Delta_1}{\Delta_2}(n-1)We\left[\left\{1+\left(We\frac{d\omega}{d\xi}\right)^2\right\}^{\frac{n-3}{2}}\right]\left(\frac{d\omega}{d\xi}\right)^2\frac{d^2\omega}{d\xi^2}+2\Omega u=0 \tag{2}$$

$$\Delta_3\Delta_4\frac{d^2\Theta}{d\xi^2}+\Delta_1 Ec\,\mathrm{Pr}\left[\left(\frac{du}{d\xi}\right)^2+\left(\frac{d\omega}{d\xi}\right)^2\right]+\frac{\Delta_2\Delta_3\Delta_4\,\mathrm{Re}}{\Delta_1}\left(A^*e^{-\xi}+B^*\Theta\right) + 4Rd\left[\left\{1+3\,(Tr-1)\,\Theta\right\}\frac{d^2\Theta}{d\xi^2}+3\,(Tr-1)\left(\frac{d\Theta}{d\xi}\right)^2\right]=0 \tag{3}$$

$$\frac{d^2\Phi}{d\xi^2}-\eta Sc\Phi^N=0 \tag{4}$$

Boundary conditions:

$$u(0)=0,\quad \omega(0)=0,\quad \Theta(0)=0,\quad \Phi(0)=0,\qquad u(1)=1,\quad \omega(1)=0,\quad \Theta(1)=1,\quad \Phi(1)=1. \tag{5}$$

Default Set of Controlling Parameters: $\Delta_1=1.0515$, $\Delta_2=1.0717$, $\Delta_3=1.0259$, $\Delta_4=1.0353$, $We=0.1$, $n=3$, $Da=0.5$, $D_f=0.5$, $\Omega=0.5$, $Ec=0.5$, $\mathrm{Pr}=3$, $\mathrm{Re}=1$, $A^*=B^*=0.3$, $Rd=1$, $\eta=0.8$, $Sc=1$, $N=2$, $Tr=1$, $P=1$.

NTK-Based Adaptive Training for PINNs

Neural Tangent Kernel (NTK):

$$K(x,x')=\left\langle\frac{\partial u(x;\theta)}{\partial\theta},\frac{\partial u(x';\theta)}{\partial\theta}\right\rangle. \tag{6}$$

For standard PINNs, the loss is

$$\mathcal{L}(\theta)=\mathcal{L}_{BC}(\theta)+\mathcal{L}_{ODE}(\theta), \tag{7}$$

and the corresponding NTK is

$$K(t)=\begin{bmatrix}K_{uu}(t) & K_{ur}(t)\\ K_{ru}(t) & K_{rr}(t)\end{bmatrix}. \tag{8}$$

where K_{uu} and K_{rr} denote the boundary and residual NTK blocks, respectively.

$$(K_{uu})_{ij}=\left\langle\frac{\partial u(x_i;\theta)}{\partial\theta},\frac{\partial u(x_j;\theta)}{\partial\theta}\right\rangle,\qquad (K_{rr})_{ij}=\left\langle\frac{\partial\mathcal{N}[u](x_i;\theta)}{\partial\theta},\frac{\partial\mathcal{N}[u](x_j;\theta)}{\partial\theta}\right\rangle. \tag{9}$$

Under gradient flow,

$$\frac{d}{dt}\begin{bmatrix}u(x_b;\theta(t))\\ \mathcal{N}[u](x_r;\theta(t))\end{bmatrix}=-K(t)\begin{bmatrix}u(x_b;\theta(t))-g(x_b)\\ \mathcal{N}[u](x_r;\theta(t))-f(x_r)\end{bmatrix}. \tag{10}$$

where x_b and x_r are the boundary and residual collocation points.

The adaptive NTK weights are

$$\lambda_b=\frac{\mathrm{Tr}(K)}{\mathrm{Tr}(K_{uu})},\qquad \lambda_r=\frac{\mathrm{Tr}(K)}{\mathrm{Tr}(K_{rr})}. \tag{11}$$

where $\mathrm{Tr}(\cdot)$ denotes the matrix trace and λ_b and λ_r are the adaptive weights assigned to the boundary-condition loss and the ODE residual loss, respectively. Accordingly, the PINN training objective becomes

$$\mathcal{L}(\theta)=\lambda_b\mathcal{L}_{BC}(\theta)+\lambda_r\mathcal{L}_{ODE}(\theta). \tag{12}$$

Algorithm 1 PINN training with adaptive NTK weighting

Input: collocation points X_f, boundary points X_b, training epochs S
Initialize network parameters θ for $\hat{u}_\theta(\xi),\hat{w}_\theta(\xi),\hat{\Theta}_\theta(\xi),\hat{\Phi}_\theta(\xi)$
Set initial weights $\lambda_{ODE}=1,\ \lambda_{BC}=1$
for $epoch=1$ to S **do**
  Evaluate network outputs at X_f and X_b
  Compute ODE residuals $r_u(\xi)$, $r_w(\xi)$, $r_\Theta(\xi)$, $r_\Phi(\xi)$
  Compute boundary residuals e_u, e_w, e_Θ, e_Φ
  Compute

$$\mathcal{L}_{ODE}=\frac{1}{N_f}\sum_{i=1}^{N_f}\left(|r_u^i|^2+|r_w^i|^2+|r_\Theta^i|^2+|r_\Phi^i|^2\right),\quad \mathcal{L}_{BC}=\frac{1}{N_b}\sum_{j=1}^{N_b}\left(|e_u^j|^2+|e_w^j|^2+|e_\Theta^j|^2+|e_\Phi^j|^2\right)$$

  if $epoch \bmod 500=0$ **then**
    Construct NTK blocks

$$K_{rr}^{(n)}=\nabla_\theta\mathcal{N}[u_\theta](X_f)\nabla_\theta\mathcal{N}[u_\theta](X_f)^T,\quad K_{uu}^{(n)}=\nabla_\theta u_\theta(X_b)\nabla_\theta u_\theta(X_b)^T$$

    Compute adaptive weights

$$\lambda_{ODE}=\frac{\mathrm{Tr}(K^{(n)})}{\mathrm{Tr}(K_{rr}^{(n)})},\quad \lambda_{BC}=\frac{\mathrm{Tr}(K^{(n)})}{\mathrm{Tr}(K_{uu}^{(n)})}$$

  end if
  Form weighted loss

$$\mathcal{L}_{total}=\lambda_{ODE}\mathcal{L}_{ODE}+\lambda_{BC}\mathcal{L}_{BC}$$

  Update network parameters by minimizing $\mathcal{L}_{total}$
end for
Output: trained solutions $\hat{u}_\theta,\hat{w}_\theta,\hat{\Theta}_\theta,\hat{\Phi}_\theta$

Results and Discussion

Performance Evaluation



Figure 2. Training Loss Curve

Table 1. Difference between collocation method and PINN-NTK with default parameter set

ξ	u	w	Θ	Φ
0	7.30E-36	0.00E+00	0.00E+00	1.66E-37
0.2	6.17E-06	1.17E-05	7.04E-06	7.41E-06
0.4	2.91E-06	1.19E-07	4.74E-06	4.60E-06
0.6	2.55E-06	1.82E-05	2.89E-06	3.54E-06
0.8	5.69E-06	1.10E-05	4.58E-06	1.25E-06
1	0.00E+00	1.68E-22	0.00E+00	0.00E+00

Limitation: The Collocation Method is a typical numeric method, requiring different initial guesses for each set of parameters, while PINN-NTK requires no initial value.

Physical Profiles

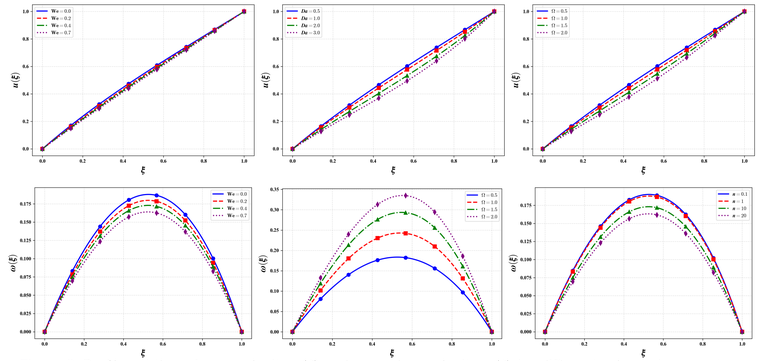


Figure 3. Profiles of the primary velocity $u(\xi)$ and secondary velocity $w(\xi)$ for different values of parameters.

Nomenclature: m (shape factor of nanoparticles), We (Weissenberg number), n (power-law index of Non-Newtonian fluid), Da (Darcy number), D_f (Forchheimer number), Ω (rotation parameter), Ec (Eckert number), Pr (Prandtl number), Re (Reynolds number), A^*, B^* (non-uniform heat source), Rd (radiation parameter), η (reaction rate parameter), Sc (Schmidt number), N (chemical reaction order), Tr (temperature ratio parameter), P (Dimensionless Pressure).

Conclusion

- An adaptive PINN-NTK framework was developed to solve the nonlinear transport problem arising in Non-Newtonian hybrid nanofluid flow through a rotating porous channel.
- The NTK-based adaptive weighting effectively balanced the ODE residual and boundary-condition losses, leading to improved training stability and more accurate predictions.
- The loss convergence and pointwise error results demonstrate that the PINN-NTK model provides reliable approximations for the primary velocity, secondary velocity, temperature, and concentration fields.
- The parametric profiles further confirm that the framework is capable of capturing the influence of the governing parameters on the flow and transport behavior.

References

[1] F. Ali, K. Loganathan, S. Eswaramoorthi, K. Prabu, A. Zaib, and D. K. Chaudhary, "Heat transfer analysis on carboxymethyl cellulose water-based cross hybrid nanofluid flow with entropy generation," *Journal of Nanomaterials*, vol. 2022, no. 1, p. 5252918, 2022.

[2] S. A. Wajihah and D. S. Sankar, "A review on non-newtonian fluid models for multi-layered blood rheology in constricted arteries," *Archive of Applied Mechanics*, vol. 93, no. 5, pp. 1771–1796, 2023.

[3] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[4] S. Wang, X. Yu, and P. Perdikaris, "When and why PINNs fail to train: A neural tangent kernel perspective," *Journal of Computational Physics*, vol. 449, p. 110768, 2022.